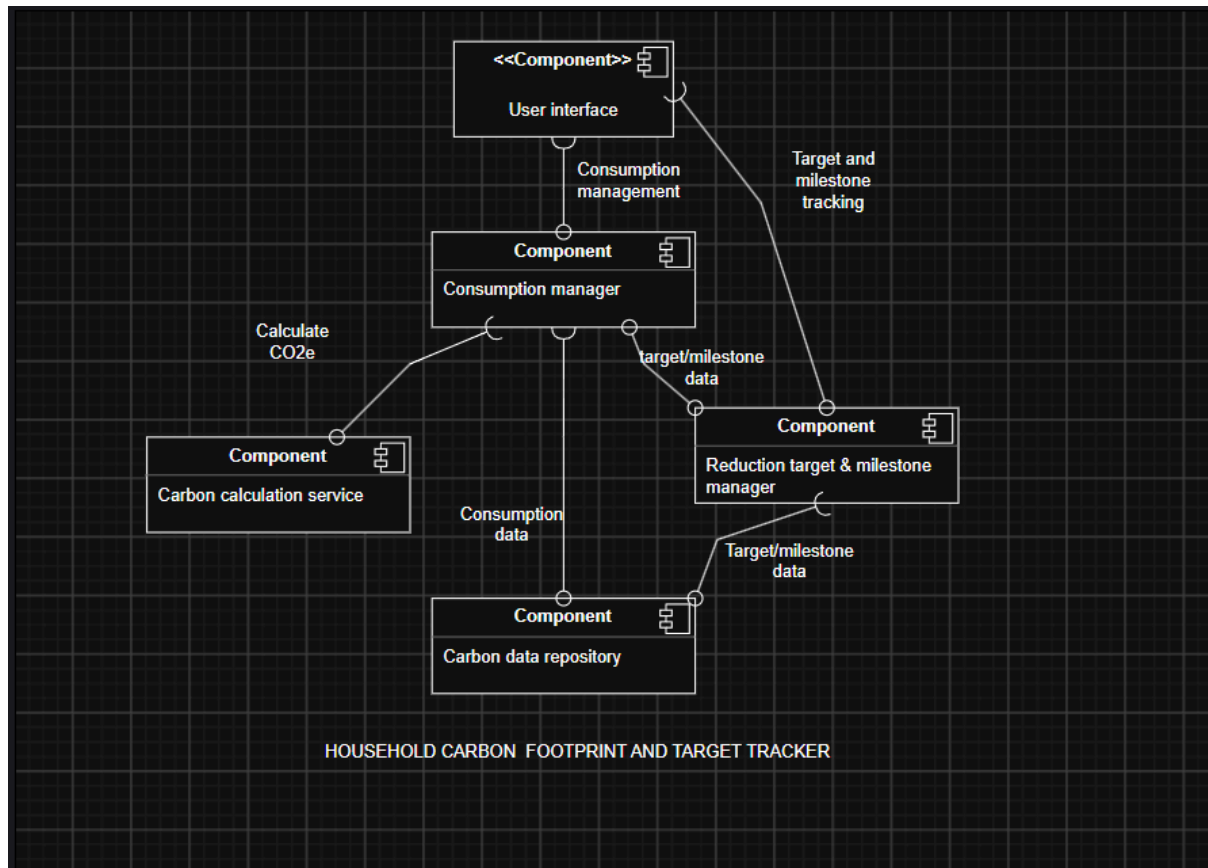


SOFTWARE ENGINEERING LAB 03
61.HOUSEHOLD CARBON FOOTPRINT & TARGET TRACKER

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UML DIAGRAM



Written Justification

Architecture Selection

The Carbon Tracker system uses a Layered Architecture. The system is divided into separate components for the user interface, consumption management, carbon calculation, reduction target and milestone management, and data storage.

Reason 1

Layered Architecture provides a clear separation between the presentation, business logic, and data management responsibilities of the Carbon Tracker system. This makes the system easier to understand, maintain, and modify because changes to one part of the system can be made without significantly affecting other parts.

Reason 2

The architecture suits the main workflow of the Carbon Tracker system. Users can enter and manage household consumption data, the system can calculate the corresponding CO₂e emissions, and users can then set and track reduction targets and milestones. Separating these functions into components makes the system structure clear and organized.

Security Advantage

Layered Architecture improves security by separating the user interface from the data repository. Users interact with the system through the appropriate management components instead of accessing stored household consumption data directly. This allows validation and access control to be handled within the application logic before data is stored or retrieved.

Performance Benefit

The separation of responsibilities allows individual components to be optimized for their specific tasks. For example, the Carbon Calculation Service can focus on performing CO₂e calculations while the Carbon Data Repository handles data storage and retrieval. This reduces unnecessary processing in the user interface and supports efficient system operation.